

Bayesian Reconstruction of Partially Observed Tongue Shapes

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Introduction

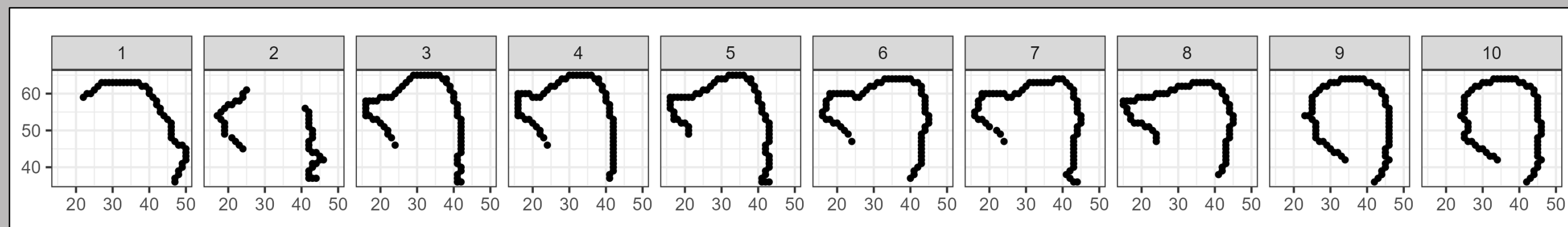


Figure 1: Curves of speaker 1.

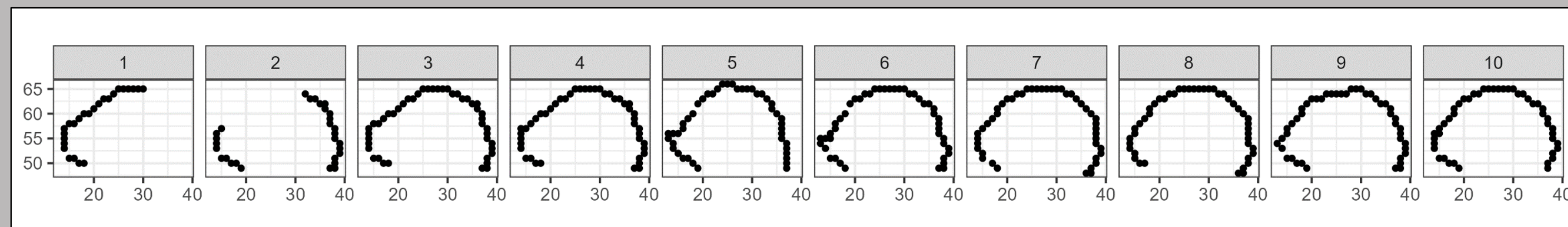


Figure 2: Curves of speaker 2.

- Tongue shapes extracted from magnetic resonance imaging (MRI)
- Two-dimensional curves: x and y coordinates

- Two speakers, 10 shapes each
- 8 fully observed and 2 partially observed

- Shapes may represent different sounds
- Speakers have different number of measurements per curve
 - 65 for speaker 1, 45 for speaker 2

General Methodology: Semiconjugate Model

- Assume a linear model where $\mathbf{a}$ is response and $\mathbf{C}$ is design matrix:

$$\mathbf{a} = \mathbf{C}\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \text{ where } \boldsymbol{\varepsilon} \sim N(\mathbf{0}, \sigma^2 \mathbf{I})$$

- Set a Normal prior for $\boldsymbol{\beta}$: $\boldsymbol{\beta} \sim N_p(\boldsymbol{\beta}_0, \boldsymbol{\Sigma}_0)$
- Set an Inverse-Gamma prior for σ^2 : $\sigma^2 \sim \text{Inv-Gamma}(\frac{\nu_0}{2}, \frac{\nu_0}{2} \sigma_0^2)$
- Approximate the joint posterior $(\boldsymbol{\beta}, \sigma^2)$ using Gibbs Sampling
 - Primary focus is on $\boldsymbol{\beta}$
- Common priors for ν_0 and σ_0^2
 - Determined from modelling x and y curve coordinates
 - Empirical distribution of $\hat{\sigma}^2$

2 Approaches

Raw Data

Coefficients

Modelling of Curve Coordinates

- Represent curve coordinates by polynomials
 - Degree 5 for x, degree 4 for y
 - Polynomials defined on $[0,1]$
 - Introduce standardised coordinate τ along curve:

$$\tau = \frac{t}{n+1}$$
 where t is the original coordinate along curve
 - Curves from speaker 2 inverted for consistency
- Orthonormal basis for polynomials
 - Normalised shifted Legendre polynomials $\ell_0, \dots, \ell_d$
- Degree d polynomials $\leftrightarrow (d+1)$ -dimensional vectors

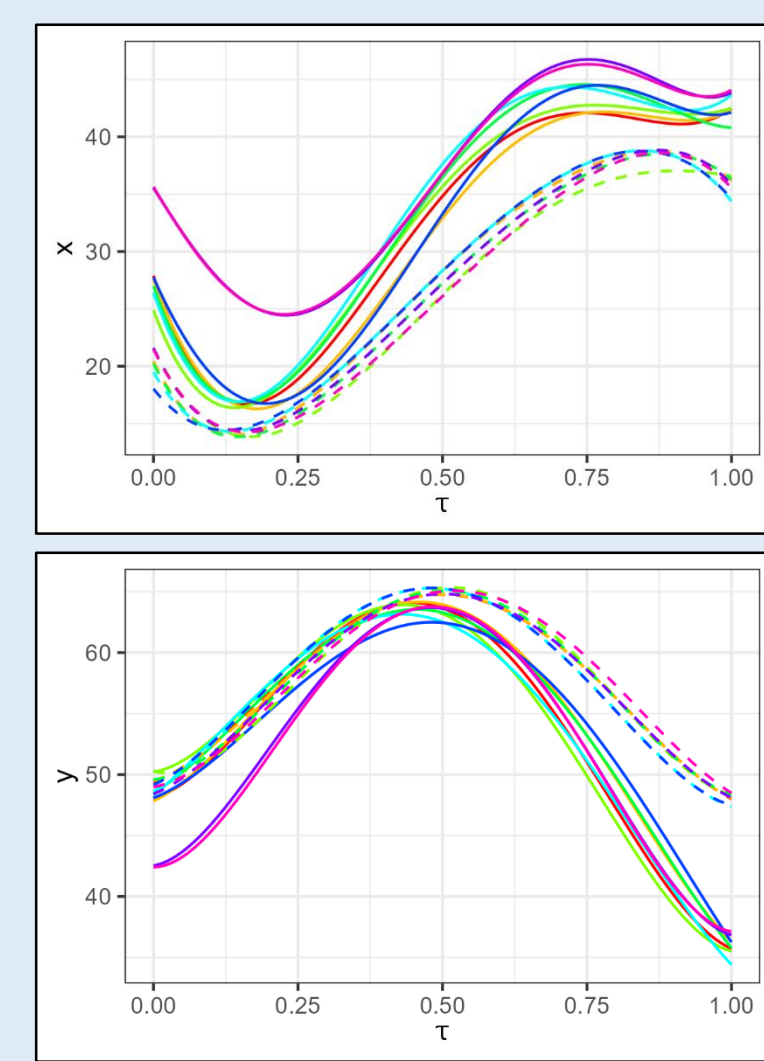


Figure 3: x (top) and y (bottom) curves. Solid line: speaker 1, dashed line: speaker 2.

No PCA

PCA

Curve Shapes

- Find $\tilde{P}_1, \dots, \tilde{P}_k$ from fully observed curves
- Model partially observed curves as a linear combination of m polynomials:

$$\Gamma = \sum_{j=1}^m \beta_j \tilde{P}_j$$

- If $m > d+1$, the polynomials are linearly dependent
- $m = d+1$ gives maximum flexibility
- Criteria for selecting the $\tilde{P}_j$ are needed

Curve Selection

- Given partially observed curve:
 - Compute sum of squared differences with each $\tilde{P}_j$
 - Order the $\tilde{P}_j$ by the result of this sum
 - Select $\lfloor \frac{m}{2} \rfloor$ best-fitting $\tilde{P}_j$
 - Select $\lfloor \frac{m}{2} \rfloor$ worst-fitting $\tilde{P}_j$

Priors

- Set prior for $\boldsymbol{\beta} \sim N_m(0.5 \cdot \mathbf{1}_m, \mathbf{I}_m)$



Main Directions of Variance

- Fit polynomials for k fully observed curves

$$\tilde{P}_j = \sum_{i=0}^d c_{ij} \ell_i$$

- Apply Principal Component Analysis (PCA)
 - PCA on vectors of coefficients $\leftrightarrow$ PCA in polynomial space
- Form a matrix with the coefficients of the observed curves in the columns

$$\begin{bmatrix} c_0^{(1)} & \dots & c_d^{(1)} \\ \vdots & \ddots & \vdots \\ c_0^{(k)} & \dots & c_d^{(k)} \end{bmatrix}$$

- Centering and scaling the vectors when doing PCA provides better accuracy
- PCA provides the coefficients of polynomials that represent the main directions of variance, $\tilde{P}_1, \dots, \tilde{P}_{d+1}$
 - Orthogonal and linearly independent
- Model partially observed curves as a linear combination:

$$\Gamma = P_0 + \sum_{j=1}^m \beta_j \tilde{P}_j$$

- No curve selection required as $m \leq d+1$ by definition ($m = d+1$ gives maximum flexibility)

Marginal Modelling

- Assumes that x and y curves are independent
- Define $\mathbf{x} = [x_1 \dots x_n]^T$ and $\mathbf{y} = [y_1 \dots y_n]^T$, where n is the number of observed curve coordinates
- Construct linear models:

$$\mathbf{x} = \mathbf{M}^{(x)} \boldsymbol{\beta}^{(x)} \text{ and } \mathbf{y} = \mathbf{M}^{(y)} \boldsymbol{\beta}^{(y)}, \text{ with}$$

$$\mathbf{M}^{(x)} = \begin{bmatrix} p_1^{(x)}(\tau_1) & \dots & p_{m_x}^{(x)}(\tau_1) \\ \vdots & \ddots & \vdots \\ p_1^{(x)}(\tau_n) & \dots & p_{m_x}^{(x)}(\tau_n) \end{bmatrix} \quad \mathbf{M}^{(y)} = \begin{bmatrix} p_1^{(y)}(\tau_1) & \dots & p_{m_y}^{(y)}(\tau_1) \\ \vdots & \ddots & \vdots \\ p_1^{(y)}(\tau_n) & \dots & p_{m_y}^{(y)}(\tau_n) \end{bmatrix}$$

where m_x and m_y denote number of polynomials

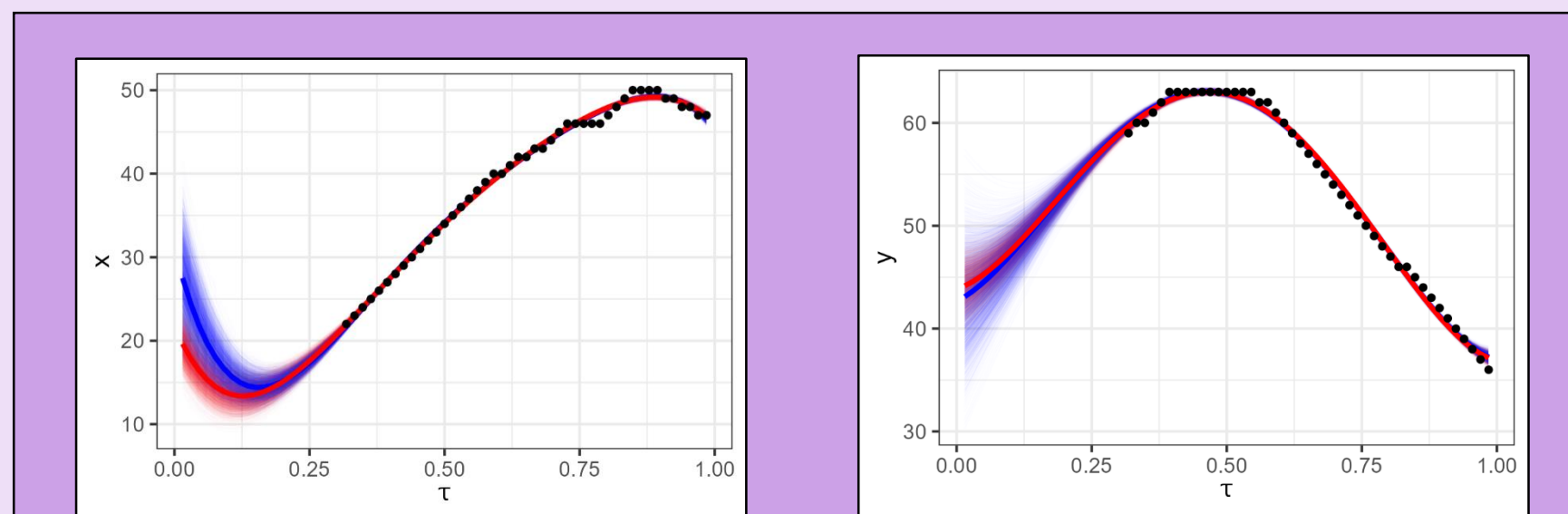


Figure 4: Results of marginal modelling for the first shape of speaker 1. Blue indicates the PCA model and red indicates the model without PCA.

Joint Modelling

- Allows for dependence between x and y curves
- Define extended response vector $\mathbf{z}$ from $\mathbf{x}$ and $\mathbf{y}$:

$$\mathbf{z} = [x_1 \dots x_n \ y_1 \dots y_n]^T$$

- Construct a stacked design matrix $\mathbf{M}^{(z)}$ from marginal $\mathbf{M}^{(x)}$ and $\mathbf{M}^{(y)}$:

$$\mathbf{M}^{(z)} = \begin{bmatrix} \mathbf{M}^{(x)} & \mathbf{0} \\ \mathbf{0} & \mathbf{M}^{(y)} \end{bmatrix}$$

- Construct linear model:

$$\mathbf{z} = \mathbf{M}^{(z)} \boldsymbol{\beta}^{(z)}, \text{ where } \boldsymbol{\beta}^{(z)} = [\boldsymbol{\beta}^{(x)} \ \boldsymbol{\beta}^{(y)}]^T$$

Priors

- Set Prior for $\boldsymbol{\beta} \sim N_m(\mathbf{0}_m, \mathbf{I}_m)$

Model Comparison

- Assess sum of absolute errors of predictions for fully observed curves based on the missing regions
 - Consider the four missing regions as defined by τ
- Models for each curve built without the corresponding shape to avoid leaking information
- Models with PCA give more stable results
- Model without PCA can perform better if not too many missing points
- Overall recommendation: **marginal PCA model**

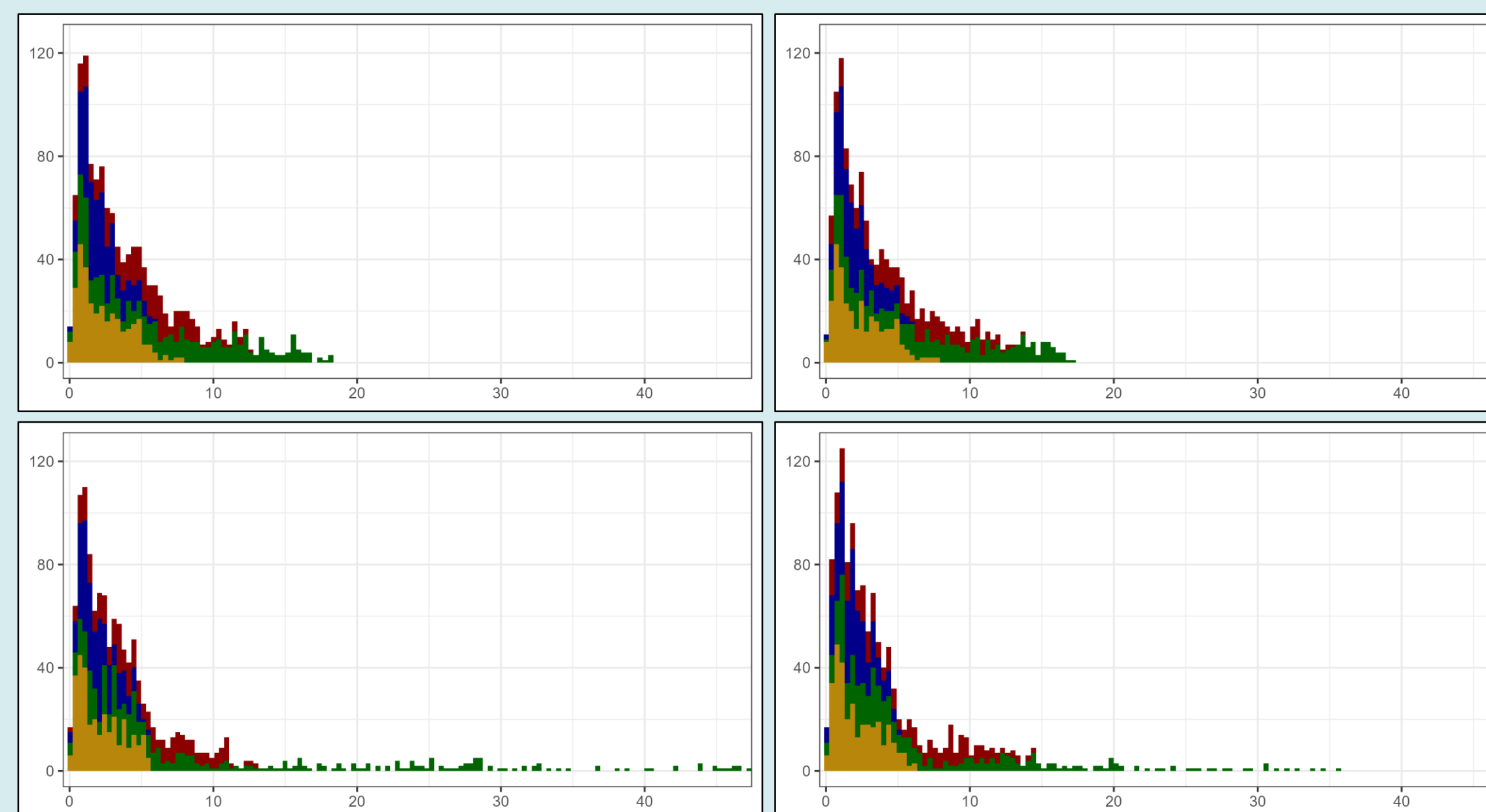


Figure 5: Histograms of pointwise absolute errors. Top left: Marginal PCA. Top right: Joint PCA. Bottom left: Marginal without PCA. Bottom right: Joint without PCA.

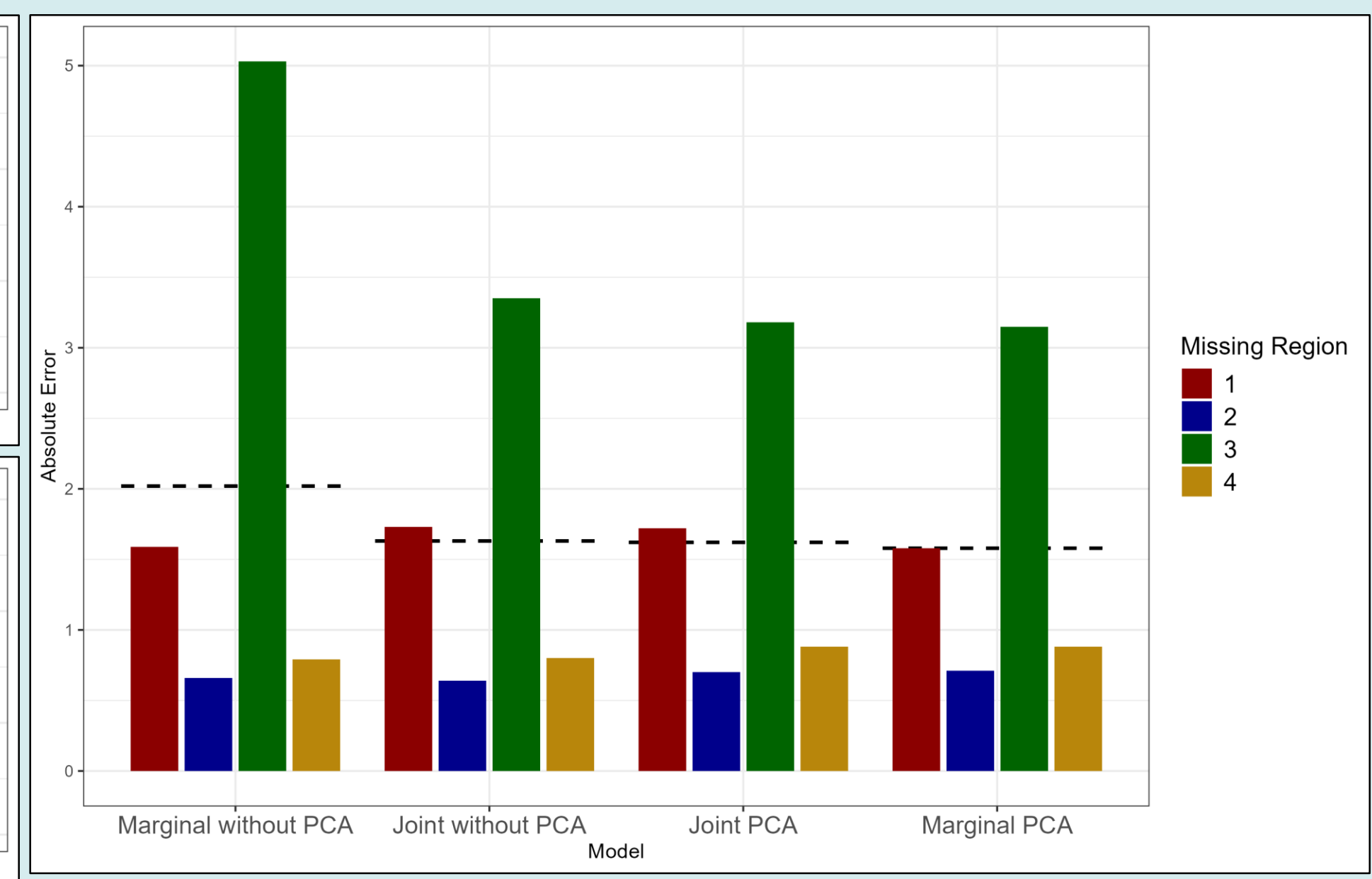


Figure 6: Average absolute error for each region and model. The black dashed line indicates the overall model average absolute error.